DESCRIPTIVE ANALYTICS

Social Media Engagement Analysis

Submitted by

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DECLARATION

I hereby declare that the project report entitled **Social Media Engagement Analysis Using Descriptive Analytics** has been prepared for academic purposes. The report demonstrates how social media post-level data can be organized, summarized, visualized, and interpreted to understand content performance across platforms. The dataset used in this report is an illustrative sample created for educational demonstration and does not represent verified performance data from an actual organization.

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ACKNOWLEDGEMENT

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ABSTRACT

This project applies descriptive analytics to social media engagement data. It examines 240 illustrative posts published across Instagram, Facebook, LinkedIn, and X during a sample twelve-month period. The analysis summarizes reach, impressions, likes, comments, shares, saves, link clicks, and engagement rate. It compares performance by platform, content category, format, and month. The sample contains 770,979 total post-level reach observations and 66,179 total engagements, with an average post engagement rate of 7.73%. These figures are calculated only from the generated educational dataset. The project demonstrates how descriptive statistics, pivot-style summaries, charts, and dashboards can support practical social media planning.

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ABBREVIATIONS

Term	Meaning
ER	Engagement Rate
CTR	Click-Through Rate
KPI	Key Performance Indicator
BI	Business Intelligence
CSV	Comma-Separated Values
ROI	Return on Investment
SMM	Social Media Marketing
impressions	Number of times content is displayed

1. TITLE AND PROJECT OVERVIEW

1.1 Project Title

Social Media Engagement Analysis Using Descriptive Analytics

1.2 Area of Study

The project relates to marketing management, digital marketing, social media marketing, consumer engagement, and business analytics. It focuses on descriptive analytics, which summarizes historical information to explain what has happened and how performance differs across categories.

1.3 Scope of the Project

• Analysis of post-level performance across four social media platforms.

• Comparison of reach, impressions, and engagement actions.

• Evaluation of content formats and content categories.

• Review of monthly reach and day-wise engagement patterns.

• Calculation of engagement rate and click-through rate.

• Presentation of findings through tables and charts.

1.4 Objectives

• Understand the role of descriptive analytics in social media marketing.

• Prepare and examine an illustrative social media dataset.

• Calculate key performance indicators for post performance.

• Compare platforms, content formats, and content categories.

• Identify high-performing content patterns in the sample.

• Suggest practical actions for improving content planning and reporting.

2. INTRODUCTION

2.1 Background of the Study

Social media platforms have become important communication and marketing channels for organizations. Businesses use them to share information, build brand awareness, engage audiences, promote products, and direct users to websites or other digital destinations. Each post generates several measurable indicators, including reach, impressions, likes, comments, shares, saves, and clicks. These indicators help marketers understand how audiences interact with published content.

2.2 Meaning of Social Media Engagement

Social media engagement refers to the interactions users make with content. Common engagement actions include likes or reactions, comments, shares, saves, and link clicks. Engagement is not identical to reach: reach estimates the number of unique accounts exposed to content, while impressions count total displays. Engagement rate relates interactions to a chosen audience or exposure base.

2.3 Meaning of Descriptive Analytics

Descriptive analytics is the process of collecting, cleaning, organizing, summarizing, and presenting historical data. It uses totals, averages, percentages, frequency distributions, cross-tabulations, and visualizations to describe patterns. It does not by itself establish causation or predict future outcomes.

2.4 Types of Data Analytics

Type	Purpose	Social Media Example
Descriptive	Explains what happened	Monthly reach and average engagement rate

Type	Purpose	Social Media Example
Diagnostic	Explores why it happened	Investigating why a campaign received fewer shares
Predictive	Estimates what may happen	Estimating future post engagement
Prescriptive	Suggests possible actions	Recommending an optimal content mix

2.5 Importance of the Study

• Helps marketers compare platform performance using consistent measures.

• Shows which content formats attract more interactions in the sample.

• Supports more informed content calendars and campaign reviews.

• Helps identify engagement gaps that require further investigation.

• Makes performance reporting clearer through visual summaries.

3. PROBLEM STATEMENT

3.1 Statement of the Problem

Organizations may publish content regularly across multiple social media platforms, but performance information can be scattered across platform dashboards and reporting periods. Looking only at follower counts or likes may provide an incomplete picture. Marketers need a structured way to compare reach, impressions, engagement actions, and clicks across platforms and content types.

3.2 Need for the Study

A descriptive analysis organizes post-level metrics into comparable summaries. It can reveal where engagement is relatively high, which content formats perform well in the sample, and how reach varies over time. Such findings can support better questions and decisions, while further testing is required before assuming that a particular format causes stronger performance.

3.3 Research Questions

• Which platform has the highest average engagement in the sample?

• Which content format has the highest average engagement rate?

• Which content category records the highest average reach?

• How does total reach vary month by month?

• Which day records the highest average engagement rate?

4. DATA DISCUSSION

4.1 Data Source and Nature

The dataset is a synthetic educational dataset containing 240 post records. It represents a hypothetical social media account portfolio over twelve sample months in 2025. It is not downloaded from platform analytics and should not be presented as actual company performance.

4.2 Dataset Structure

Field	Description
Post_ID	Unique identifier for each post
Month	Month of publication
Platform	Instagram, Facebook, LinkedIn, or X
Content_Category	Educational, Promotional, Behind the Scenes, Testimonial, or Festive

Field	Description
Content_Format	Reel/Video, Carousel, Static Image, or Story
Day	Day of publication
Followers	Illustrative follower base at posting
Reach / Impressions	Estimated unique reach and total displays
Likes, Comments, Shares, Saves	Engagement action counts
Link_Clicks	Illustrative link-click count
Total_Engagement	Sum of likes, comments, shares, and saves
Engagement_Rate_%	Total engagement divided by reach × 100
Click_Through_Rate_%	Link clicks divided by impressions × 100

4.3 Key Performance Indicators

Engagement Rate (%) = (Likes + Comments + Shares + Saves) ÷ Reach × 100

Click-Through Rate (%) = Link Clicks ÷ Impressions × 100

Average Reach = Total reach across posts ÷ Number of posts

Total Engagement = Likes + Comments + Shares + Saves

4.4 Data Quality Considerations

• Each post has a unique Post_ID.

• Numeric metrics are stored as numbers for calculations.

• Platform, category, format, and day values use consistent labels.

• Engagement rate and click-through rate are calculated from the underlying counts.

• The synthetic nature of the dataset is clearly disclosed to avoid misrepresenting the results.

5. METHOD, TOOLS AND TECHNOLOGIES USED

5.1 Research Methodology

The project follows a quantitative, descriptive approach. The process begins with preparing a structured dataset, checking field consistency, calculating performance indicators, grouping records by relevant dimensions, and presenting results through tables and charts. The analysis describes the sample rather than testing causal relationships.

5.2 Tools and Technologies

Tool	Application
Microsoft Excel	Data entry, cleaning, formulas, sorting, filtering, pivot tables, and charts
Power BI	Conceptual dashboard design and interactive reporting
Descriptive Statistics	Totals, averages, percentages, and frequency summaries
Charts and Tables	Visual comparison of platforms, formats, categories, and months

5.3 Analytical Techniques

• Frequency distribution to count posts by platform and format.

• Sum and average calculations for reach and engagement.

• Percentage calculations for engagement rate and click-through rate.

• Grouping and comparison across platform, category, format, month, and day.

• Column charts and line charts for visual interpretation.

6. WORKFLOW

Step	Activity	Output
1	Define project objectives	Analysis questions
2	Create structured sample data	240 post records
3	Check and clean fields	Consistent dataset
4	Calculate KPIs	Engagement and click rates
5	Group and summarize	Platform/category/month summaries
6	Visualize results	Charts and tables
7	Interpret findings	Observations
8	Recommend actions	Practical marketing suggestions

6.1 Workflow Explanation

The workflow ensures that data preparation is completed before interpretation. After validation, the dataset is summarized using consistent measures. Visualizations are then used to make comparisons easier to understand. Finally, findings are converted into recommendations while keeping the limitations of the sample in view.

7. PROCEDURE

7.1 Data Preparation

The 240 post records were arranged in a tabular structure. Each row represents one post and each column represents a descriptive attribute or performance metric. Unique identifiers were assigned to avoid duplicate post references.

7.2 Data Cleaning

• Checked for blank values in important fields.

• Standardized platform and content labels.

• Verified that numeric columns contained numeric values.

• Reviewed calculated engagement totals against component interactions.

• Calculated engagement and click-through rates using consistent denominators.

7.3 Data Analysis

The dataset was grouped by platform, content format, content category, and month. Counts and averages were calculated to compare performance. The average engagement rate was used to compare relative interaction levels, while average reach was used to compare exposure.

7.4 Visualization

Bar charts were selected for category comparisons, while a line chart was used to display monthly reach. Tables provide exact summary values and support interpretation.

8. RESULTS AND ANALYSIS

8.1 Overall KPI Summary

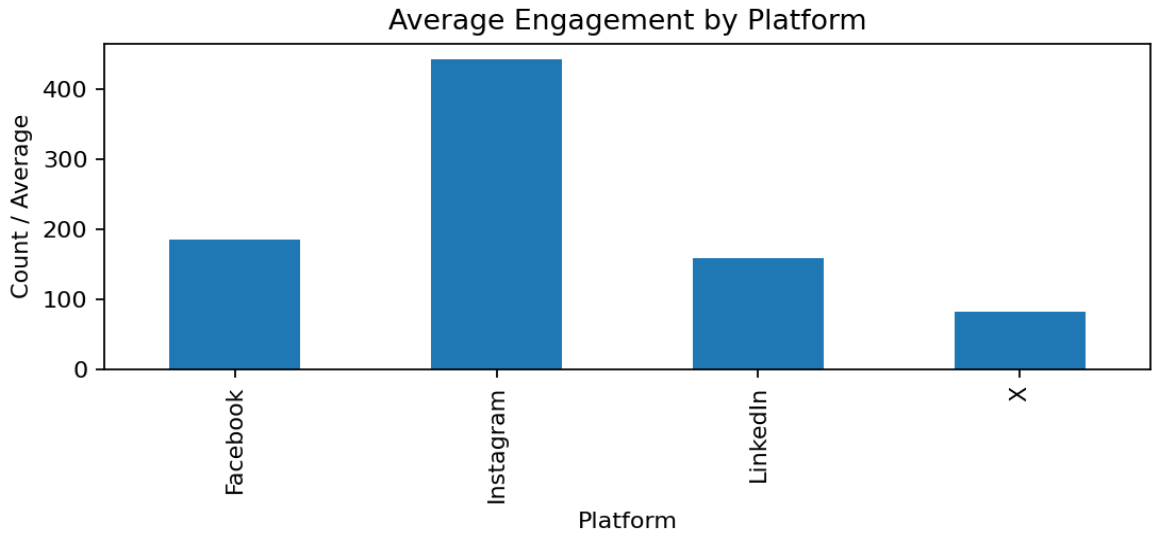
Metric	Value
Total posts	240
Total reach (sum of post reach)	770,979

Metric	Value
Total engagements	66,179
Average reach per post	3,212
Average engagement rate	7.73%
Highest average engagement platform	Instagram
Highest average engagement-rate format	Reel/Video
Highest average-reach category	Festive
Highest average engagement-rate day	Sunday

8.2 Platform-wise Analysis

Platform	Posts	Avg. Reach	Avg. Engagement	Avg. ER %
Facebook	62	2,580	186	6.67
Instagram	99	4,649	443	9.25
LinkedIn	57	2,174	159	6.91
X	22	1,223	82	5.98

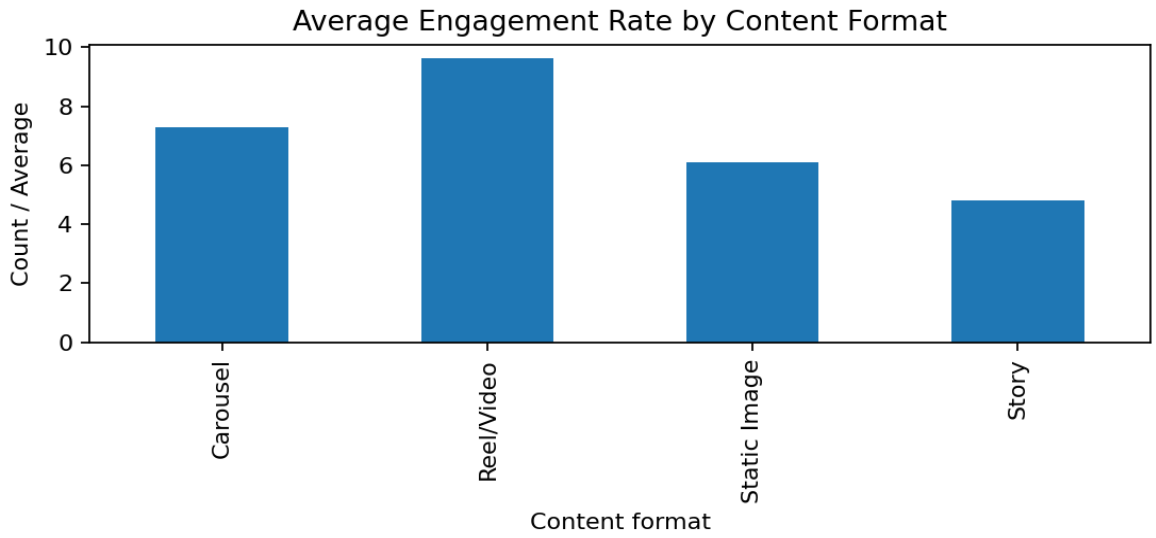
The platform comparison shows that **Instagram** has the highest average engagement per post in this illustrative dataset. This should be interpreted alongside average reach and the number of posts, since platform audience sizes and content distribution differ.



8.3 Content Format Analysis

Format	Posts	Avg. Reach	Avg. ER %
Carousel	66	2,968	7.28
Reel/Video	98	4,128	9.61
Static Image	52	2,473	6.11
Story	24	1,747	4.81

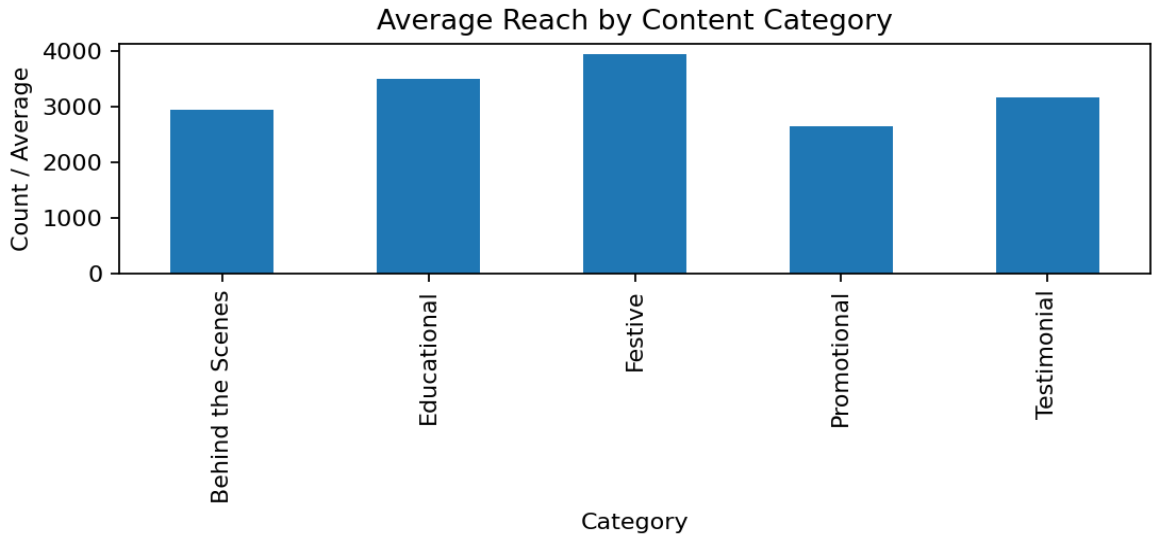
The highest average engagement rate is recorded by **Reel/Video**. The result suggests that format-level performance deserves attention when planning content. However, a fair comparison should also consider content quality, publishing time, audience size, and campaign objectives.



8.4 Content Category Analysis

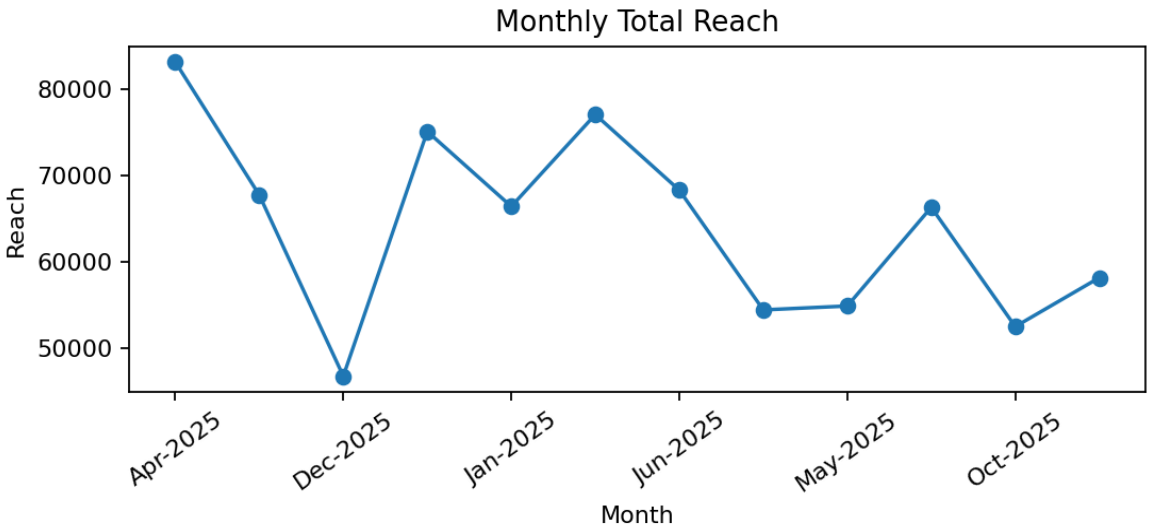
Category	Posts	Avg. Reach	Avg. ER %
Behind the Scenes	55	2,938	7.18
Educational	38	3,497	7.93
Festive	48	3,941	7.92
Promotional	51	2,652	7.90
Testimonial	48	3,168	7.84

The category with the highest average reach is **Festive**. Category-level comparisons can help marketers balance informative, promotional, social-proof, and seasonal content rather than relying on a single content type.



8.5 Monthly Reach Analysis

Monthly reach is calculated by adding the reach values of all posts published in each month. It represents the sum of post-level reach observations and may include the same account being reached by multiple posts; it should not be interpreted as unique annual audience size.



8.6 Engagement and Click Indicators

The average engagement rate across posts is **7.73%**. Engagement rate provides a normalized view of interactions relative to reach. Click-through rate is calculated using link clicks divided by impressions and is useful when the objective is to encourage users to visit an external destination.

8.7 Interpretation of Results

• Instagram leads the sample in average total engagement per post.

• The strongest format by average engagement rate is Reel/Video.

• The highest average reach by content category is Festive.

• Monthly reach totals vary, indicating that publishing volume and post performance should be reviewed together.

• The best-performing sample category or format should be tested consistently before being adopted as a universal rule.

9. OBSERVATIONS

• The dataset contains 240 illustrative social media posts across four platforms.

• Platform comparisons show differences in average reach and engagement.

• Engagement rate provides a more comparable measure than raw engagement totals alone.

• Content format and category are useful dimensions for organizing performance reviews.

• Monthly totals help identify periods of relatively higher and lower post-level reach.

• The analysis is descriptive and cannot establish that platform or format alone caused the observed results.

10. RECOMMENDATIONS

• Develop platform-specific content plans instead of publishing identical content everywhere.

• Use the stronger-performing formats as test candidates, while monitoring reach, saves, shares, and comments separately.

• Maintain a balanced content mix of educational, promotional, testimonial, behind-the-scenes, and seasonal posts.

• Track engagement rate and click-through rate alongside reach and impressions.

• Review performance monthly and compare like-for-like campaigns where possible.

• Use consistent naming conventions and maintain a central reporting sheet.

• Run controlled experiments before making causal claims about posting time, format, or creative style.

• Set objectives for each post, such as awareness, engagement, traffic, or lead generation.

11. CONCLUSION

The project demonstrates how descriptive analytics can turn social media post records into structured marketing information. By calculating and comparing reach, impressions, engagement actions, engagement rate, and click-through rate, the analysis provides a clear view of the illustrative dataset. Platform-wise, format-wise, category-wise, and monthly summaries highlight differences that can support content planning and performance reporting. The findings are limited to the synthetic sample and should be treated as an academic demonstration rather than actual platform benchmarks. Overall, descriptive analytics provides a practical foundation for understanding social media performance and asking better marketing questions.

12. LIMITATIONS AND FUTURE SCOPE

12.1 Limitations

• The dataset is synthetic and does not represent a real brand or account.

• Only 240 post records are included.

• Reach and engagement values are illustrative rather than platform-verified.

• The analysis does not control for paid promotion, audience demographics, campaign objectives, or algorithm changes.

• The results describe associations and averages; they do not prove causation.

12.2 Future Scope

• Use actual platform-exported analytics with appropriate permission.

• Extend the study to multiple years and larger datasets.

• Add audience demographics, campaign spend, conversions, and lead quality.

• Compare organic and paid performance separately.

• Use diagnostic or predictive analytics in a separate project to investigate drivers or estimate future outcomes.

13. REFERENCES

• Microsoft Support – Excel help and documentation: https://support.microsoft.com/excel

• Microsoft Learn – Power BI documentation: https://learn.microsoft.com/power-bi/

• Meta Business Help Center – Insights and measurement resources: https://www.facebook.com/business/help

• LinkedIn Marketing Solutions – Marketing and analytics resources: https://business.linkedin.com/marketing-solutions

• X Business – Analytics and advertising resources: https://business.x.com/

Note: References are listed as general learning resources. The numerical dataset and calculations in this report were generated for academic demonstration.

14. APPENDIX: SAMPLE DATASET

The following rows provide a small preview of the 240-record illustrative dataset. The complete calculations and analysis in this PDF use the full dataset.

Post ID	Month	Platform	Category	Format	Reach	Likes	Comments	Shares	Saves	Clicks	ER %
P001	May-2025	Facebook	Behind the Scenes	Story	1177	31	13	6	15	21	5.52
P002	Jan-2025	LinkedIn	Promotional	Story	1898	42	11	11	19	18	4.37
P003	Oct-2025	Instagram	Testimonial	Static Image	2538	61	22	9	41	28	5.24
P004	Sep-2025	LinkedIn	Testimonial	Reel/Video	293	23	2	2	4	2	10.58

Post ID	Month	Platform	Category	Format	Reach	Likes	Comments	Shares	Saves	Clicks	ER %
P005	Jun-2025	Instagram	Behind the Scenes	Reel/Video	4180	148	41	29	12	30	5.50
P006	Oct-2025	Instagram	Testimonial	Reel/Video	12281	1050	69	45	70	77	10.05
P007	Sep-2025	Instagram	Promotional	Reel/Video	6442	653	54	60	44	29	12.59
P008	May-2025	Instagram	Promotional	Reel/Video	2878	153	14	25	7	37	6.91
P009	Apr-2025	Instagram	Testimonial	Reel/Video	7714	482	57	97	79	127	9.27
P010	Jun-2025	Instagram	Promotional	Reel/Video	7470	629	55	79	56	95	10.96
P011	Aug-2025	Facebook	Festive	Carousel	4377	267	22	17	56	80	8.27
P012	May-2025	Instagram	Testimonial	Static Image	2106	152	21	2	16	12	9.07

Dataset note: The full 240-row dataset is not embedded in this PDF. The appendix shows a preview only. All findings are calculated from the complete internally generated dataset.